Probability Distributions Part 1

- Random Variables and their Probability Distributions
- Expected Values of Random Variables
- The Binomial Distribution
- The Poisson Distribution

SESSION OBJECTIVES:

- ☑ Distinguish between discrete random variables and continuous random variables
- ☑ Describe and provide examples of both discrete and continuous random variables

SESSION OBJECTIVES:

- ☑ Calculate the expected value (mean), variance, and standard deviation of a discrete random variable
- ☑ Explain the properties of a binomial and Poisson distribution

SESSION OBJECTIVES:

- ☑ Apply the binomial distribution to applied problems
- ☑ Compute the probabilities for the Poisson distributions

Random Variables and their Probability Distributions

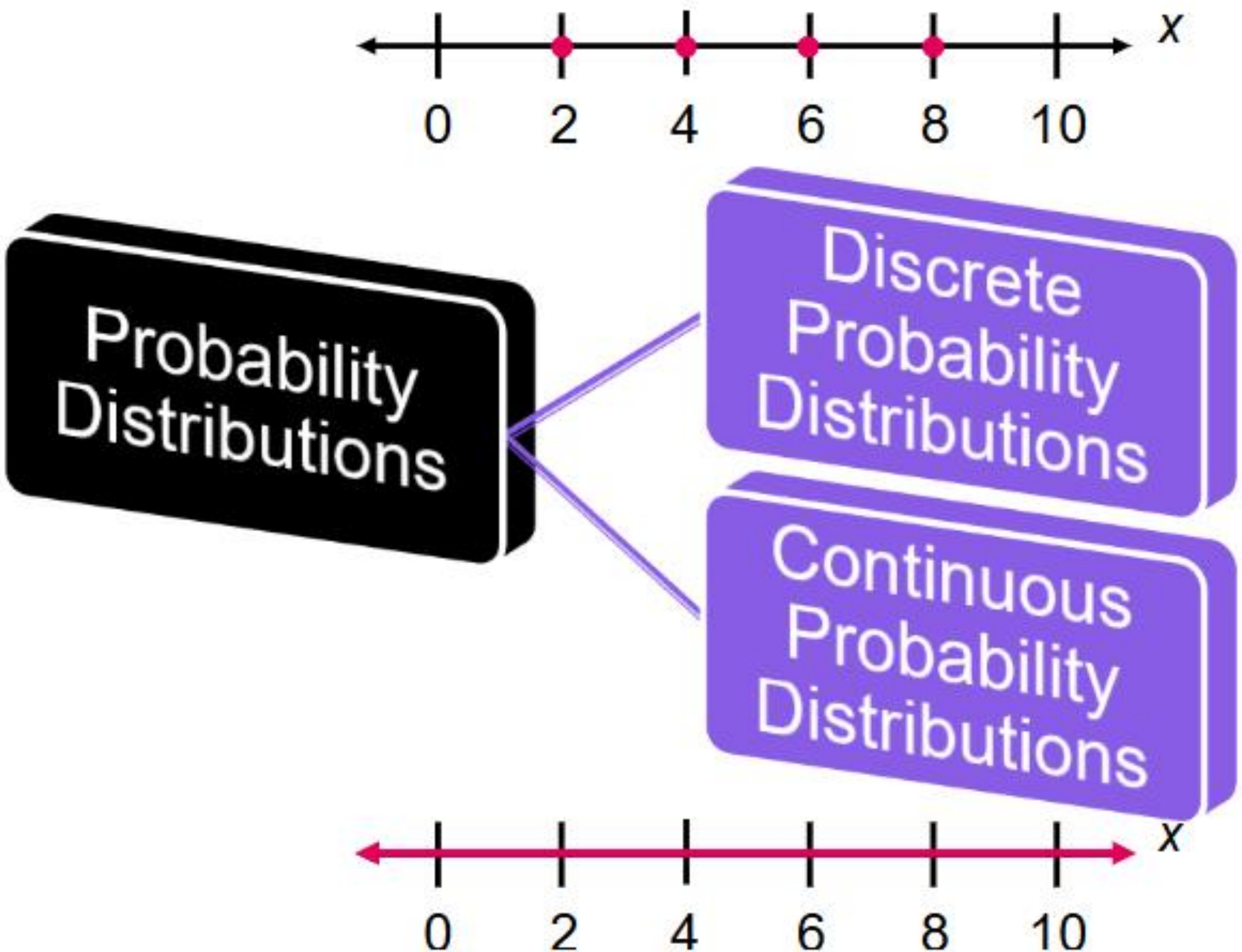
WHAT IS A RANDOM VARIABLE?

$$P(X = x)$$

Random Variable

Random Variable (a.k.a. chance variable, a stochastic variable, or simply a variate) is a function whose domain is a sample space and whose range is some set of real numbers. It is usually denoted by capital or uppercase letters of the English alphabet (e.g., X), and the possible values of the variables are usually denoted by corresponding lowercase letters (e.g., x)

Random Variables and their Probability Distributions



Discrete Random Variable is a random variable that may assume a finite or $\boxed{\text{countable number}}$ of possible outcomes that can be listed.

Examples:

- Number of siblings of an individual
- Number of rings before the phone is answered
- Number of students who were protesting the tuition increase last semester
- Number of accidents at a certain intersection over one (1) years' time

Continuous Random Variable is a random variable that may assume an $\boxed{\text{uncountable number}}$ of values or possible outcomes, represented by the intervals on a number line.

Examples:

- The concentration of pollutants in a local water source
- The depth of drilling to find oil

SELF CHECK:

Discrete Random Variable or Continuous Random Variable?

DISCRETE — The number of cats in a shelter at any given time

CONTINUOUS — The weight of newborn babies

CONTINUOUS — The weight of a book in the library

DISCRETE — The types of book in the library

SELF CHECK:

Discrete Random Variable or Continuous Random Variable?

DISCRETE — The number of books in the library

CONTINUOUS — The average number of stars a business is rated online

DISCRETE — The grade given to a student, as a letter

CONTINUOUS — The grade given to a student, as a percentage

Random Variables and their Probability Distributions

WHAT IS A PROBABILITY MASS FUNCTION?

Probability Mass Function (pmf)

Probability Mass Function (pmf) provides the probabilities $f(x) = P(X = x)$ for all possible values that a discrete random variable (x) can take on in the range of X. This function may be viewed or can be represented as a table, graph, or formula.

Simple Explanation of **Probability Mass Function (PMF)**

A **Probability Mass Function (PMF)** is used in **discrete probability distributions**. It tells us the probability that a **discrete random variable** is exactly equal to some value.

In simple terms:

- A PMF gives the probability of each possible **specific outcome**.
- It only applies to **discrete** random variables (i.e., variables that take **countable values** like 0, 1, 2...).

Random Variables and their Probability Distributions

WHAT IS A PROBABILITY DISTRIBUTION?

Probability Distribution

Probability Distribution is a function that describes the shape, character, and relative likelihoods of obtaining the possible values that a random variable can assume.

Simple Explanation of **Probability Distribution**

A **probability distribution** tells us **how likely** each outcome is for a random variable. It shows **all possible values** a variable can take and the **probability** of each.

There are two main types:

1. **Discrete Probability Distribution** – for countable values (like number of heads when flipping a coin).
2. **Continuous Probability Distribution** – for values over an interval (like height or temperature).

Random Variables and their Probability Distributions

Example 1:

- A wallet contains four (4) ₱100 bills, two (2) ₱200 bills, three (3) ₱500 bills, and one (1) ₱1,000 bill. Construct a probability distribution for the data.

Source: www.bsp.gov.ph

Solution:

- Step 1: List all possible outcomes. Let the variable X represent the possible totals, and the events are described as follows:
  - E_1 is for ₱100 bills (₱100 bill has 4 pieces; thus, $n(E_1)$ = 4)
  - E_2 is for ₱200 bills (₱200 bill has 2 pieces; thus, $n(E_2)$ = 2)
  - E_3 is for ₱500 bills (₱500 bill has 3 pieces; thus, $n(E_3)$ = 3)
  - E_4 is for ₱1,000 bill (₱1,000 bill has 1 piece; thus, $n(E_4)$ = 1).

$n(E_1)$	+	$n(E_2)$	+	$n(E_3)$	+	$n(E_4)$	=	Possible Total $n(E) = X$
4	+	**2**	+	**3**	+	**1**	**=**	**10**

- Step 2: Identify the outcomes that represent the same event. Find the probability of each event. Thus, the probability $P(x)$ can be calculated for each x by dividing the number of particular bills by the total number of bills.

- For ₱100 bills:
- $F(4) = P(x = 4) = f(4) = \frac{4}{10} = 0.40$

- For ₱200 bills:
- $F(2) = P(x = 2) = f(2) = \frac{2}{10} = 0.20$

- For ₱500 bills:
- $F(3) = P(x = 3) = f(3) = \frac{3}{10} = 0.30$

- For ₱1,000 bills:
- $F(1) = P(x = 1) = f(1) = \frac{1}{10} = 0.10$

- Step 3: Make a table or a histogram in which one column (or x-axis) represents the events, and the other column (or y-axis) represents the probability. The probability distribution is shown below:

Number of bills X	₱100	₱200	₱500	₱1,000
Probability P(x)	0.40	0.20	0.30	0.10

Random Variables and their Probability Distributions

Example 2:

- Find the probability distribution of the sum of the numbers when a pair of dice is tossed.

Solution:

Step 1: List all possible outcomes.

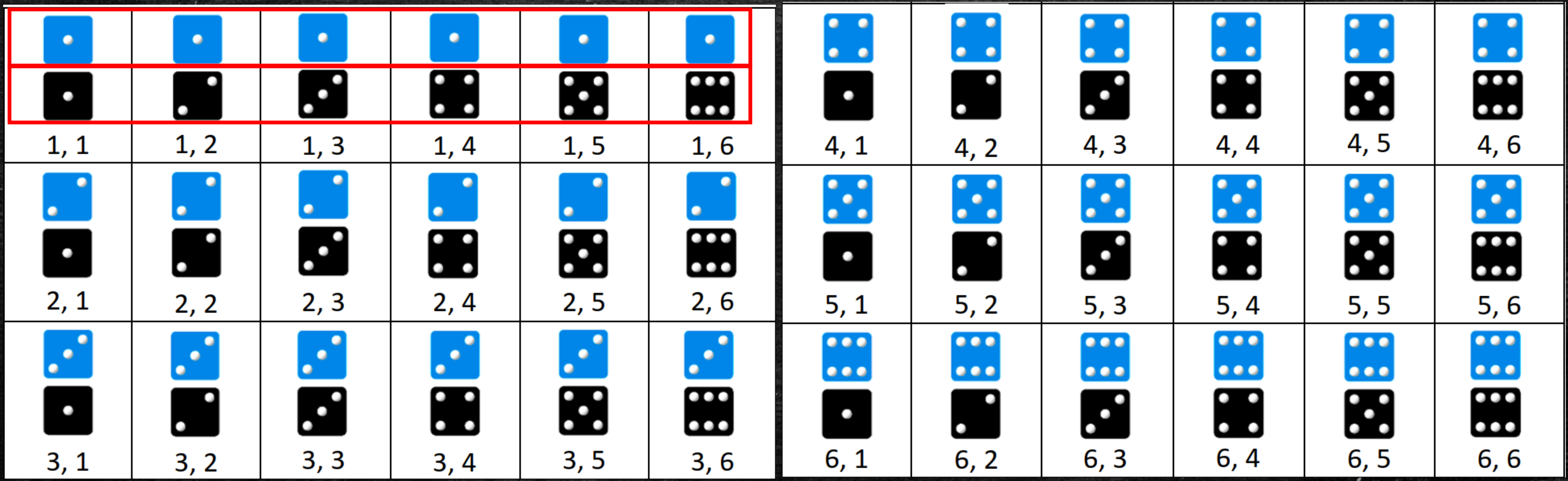
Let the variable X represent the possible totals, and the events are described as follows:

- E_1 is for the 1st die (The die can land in 6 ways (1, 2, 3, 4, 5, 6), so we can have $n(E_1) = 6$).
- E_2 is for the 2nd die (The die can land in 6 ways (1, 2, 3, 4, 5, 6), so we can have $n(E_2) = 6$).

Thus, the possible total (X) is:

# of ways a 1st dice can land $n(E_1)$	$\times$	# of ways a 2nd dice can land $n(E_2)$	=	Possible Total $n(E) = X$
6	$\times$	**6**	**=**	**36**

There are 36 possible results. Thus, the sample space has 36 elements as shown below.



Step 2: Identify the outcomes that represent the same event. Find the probability of each event.

What are the possible values of X?

The possible values for *X* are the numbers 2 through 12 as shown below:

1+1 = **2**	1+2 = **3**	1+3 = **4**	1+4 = **5**	1+5 = **6**	1+6 = **7**
2+1 = **3**	2+2 = **4**	2+3 = **5**	2+4 = **6**	2+5 = **7**	2+6 = **8**
3+1 = **4**	3+2 = **5**	3+3 = **6**	3+4 = **7**	3+5 = **8**	3+6 = **9**
4+1 = **5**	4+2 = **6**	4+3 = **7**	4+4 = **8**	4+5 = **9**	4+6 = **10**
5+1 = **6**	5+2 = **7**	5+3 = **8**	5+4 = **9**	5+5 = **10**	5+6 = **11**
6+1 = **7**	6+2 = **8**	6+3 = **9**	6+4 = **10**	6+5 = **11**	6+6 = **12**

Table 2. All Possible Sums of Two (2) Dice

Below are the values of *X* associated with some of the outcomes on *Tables 1* and *2*.

Value of *X*	**Number of outcomes**		**Description**
2	1	(1, 1)	If *X* = 2, the number showing on the first die must be 1 and the second die also is 1. There is only one (1) chance out of 36 that both dice show 1.
3	2	(2, 1) (1, 2)	When *X* = 3, the first die shows 1 and the second die shows 2, or vice versa. Thus, there are two (2) chances in 36 of this happening.
4	3	(1, 3) (2, 2) (3, 1)	There are three (3) possible outcomes for a sum of 4 or 10.

5	4	(1, 4) (2, 3) (3, 2) (4, 1)	There are four possible outcomes for a sum of 5 or 9.
6	5	(1, 5) (2, 4) (3, 3) (4, 2) (5, 1)	There are five (5) outcomes for a sum of 6 or 8.
7	6	(1, 6) (2, 5) (3, 4) (4, 3) (5, 2) (6, 1)	There are six (6) possible outcomes for the sum of 7.
8	5	(2, 6) (3, 5) (4, 4) (5, 3) (6, 2)	There are five (5) outcomes for a sum of 6 or 8.

9	4	(3, 6) (4, 5) (5, 4) (6, 3)	There are four (4) possible outcomes for a sum of 5 or 9.
10	3	(4, 6) (5, 5) (4, 6)	There are three (3) possible outcomes for a sum of 4 or 10.
11	2	(5, 6) (6, 5)	There are two (2) possible outcomes for a sum of 3 or 11.
12	1	(6, 6)	There is only one (1) possible outcome for the sum of 2 or 12.

Step 3: Make a table or a histogram in which one (1) column (or x-axis) represents the events, and the other column (or y-axis) represents the probability.

Note that both dice are fair and the rolls are independent. Thus, the probability mass function of X is computed as follows:

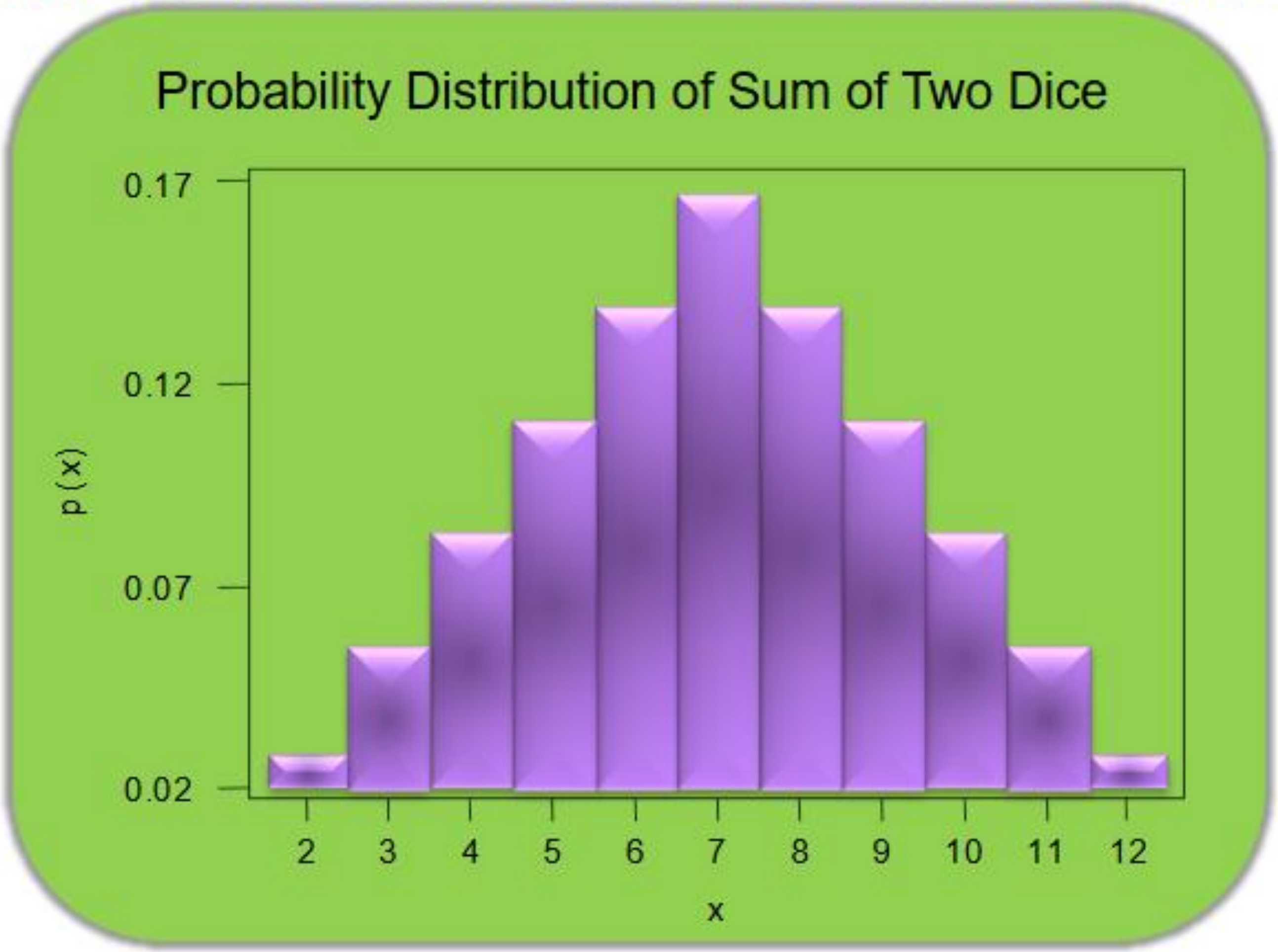
Description	
The event symbolized by $x = 1$ is the null event of the sample space Ω, since the sum of the numbers on the dice cannot be at most 1.	$F(1) = P(x = 1) = 0$
$x = 2$ is the event $\{(1,1)\}$	Thus, $F(2) = P(x = 2) = f(2) = \frac{1}{36}$
$x = 3$ is the event $\{(1,2),(2,1)\}$	Thus, $F(3) = P(x = 3) = f(3) = \frac{2}{36}$

The probabilities $f(2), f(3), \ldots, f(12)$ in an analogous manner. Continuing this way, we obtain the table below:

Sum (*X*)	2	3	4	5	6	7	8	9	10	11	12
Probability P(*X* = *x*)	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$
Reduced Probability	$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{5}{36}$	$\frac{1}{6}$	$\frac{5}{36}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{18}$	$\frac{1}{36}$

The table only includes those numbers x for which $f(x) > 0$. And since we include all such numbers, the probabilities f(x) in the table add to 1.

Random Variables and their Probability Distributions



* Note that: $P(x) = (6 - \sqrt{(7-x)^2})/36$

Random Variables and their Probability Distributions

Example 3:

- Construct a probability distribution for the number of heads when three (3) balanced coins are tossed.

Solution:

Step 1: List all possible outcomes.

Let the variable x represent the possible totals, and the events are described as follows:

- E_1 is for the 1st coin (The coin can land in two [2] ways (head (H), tail (T)), so we can have $n(E_1) = 2$).
- E_2 is for the 2nd coin (The coin can land in two [2] ways (head (H), tail (T)), so we can have $n(E_2) = 2$).
- E_3 is for the 3rd coin (The coin can land in two [2] ways (head (H), tail (T)), so we can have $n(E_3) = 2$).

Thus, the possible total (X) is:

# of ways a 1st coin can land $n(E_1)$	$\times$	# of ways a 2nd coin can land $n(E_2)$	$\times$	# of ways a 3rd coin can land $n(E_3)$	=	Possible Total $n(E) = X$
2	$\times$	**2**	$\times$	**2**	**=**	**8**

Since there are three (3) coins, there are a total of eight (8) outcomes. Thus, the sample space has 24 elements as shown below.

1st coin	2nd coin	3rd coin	1st coin	2nd coin	3rd coin

Step 2: Identify the outcomes that represent the same event. Find the probability of each event.

Since we are interested in the number of heads that occur, there would be four (4) events. Below are the values of *X* associated with some of the outcomes on *Table 3*.

1^{st} Coin	2^{nd} Coin	3^{rd} Coin	Number of heads (X)
Head (H)	Head (H)	Head (H)	Three (3) heads, zero tails
Head (H)	Head (H)	Tail (T)	Two (2) heads, one (1) tail
Head (H)	Tail (T)	Head (H)	
Tail (T)	Head (H)	Head (H)	
Head (H)	Tail (T)	Tail (T)	One (1) head, two (2) tails
Tail (T)	Head (H)	Tail (T)	
Tail (T)	Tail (T)	Head (H)	
Tail (T)	Tail (T)	Tail (T)	Zero heads, three (3) tails

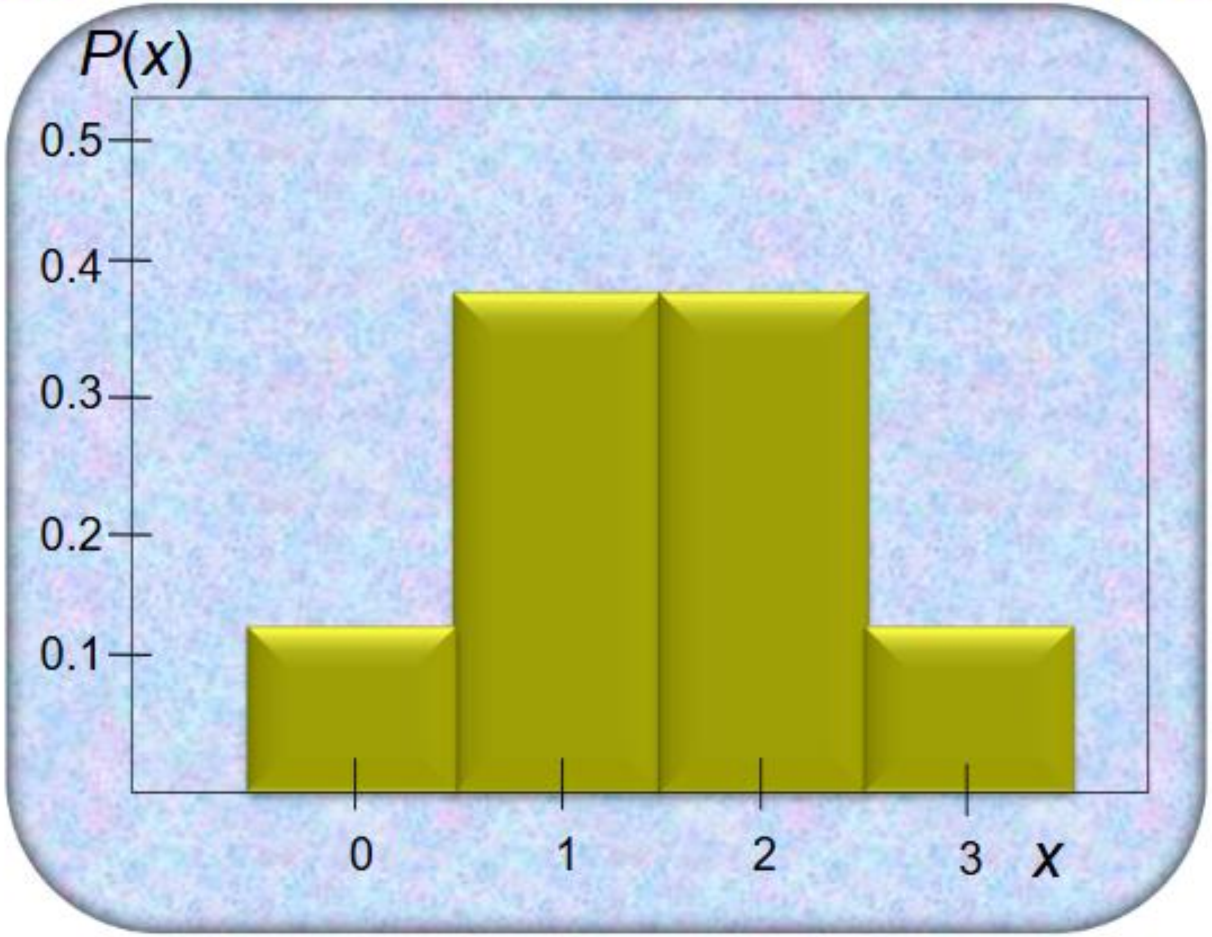
- Step 3: Make a table or a histogram in which one column (or x-axis) represents the events, and the other column (or y-axis) represents the probability. Thus, the probability mass function of X is computed as:

Description	
$x = 3$ is the event $\{(H,H,H)\}$	Thus, $F(3) = P(x = 3) = f(3) = \frac{1}{8}$
$x = 2$ is the event $\{(H,H,T),(H,T,H),(T,H,H)\}$	Thus, $F(2) = P(x = 2) = f(2) = \frac{3}{8}$
$x = 1$ is the event $\{(H,T,T),(T,H,T),(T,T,H)\}$	Thus, $F(1) = P(x = 1) = f(1) = \frac{3}{8}$
$x = 0$ is the event $\{(T,T,T)\}$	Thus, $F(0) = P(x = 0) = f(0) = \frac{1}{8}$

From the result above, a probability distribution can be constructed by listing the outcomes and assigning the probability of each outcome, as follows.

Number of heads X	0	1	2	3
Probability P(x)	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Random Variables and their Probability Distributions



SELF CHECK:

Solve

Practice Exercise 1

of 03 Practice Exercises 1

03 PE1

Instruction: Solve the following problems on a clean sheet of paper. Show the complete solution. (3 items x 20 points)

1. In a class of 100 students, 80 students passed in all subjects, 10 failed in one subject, 7 failed in two subjects and 3 failed in three subjects. Find the probability distribution of the variable for number of subjects a student from the given class has failed in. Create a probability distribution in table and a histogram.
2. A card is drawn at random from a standard deck. What is the probability distribution for drawing an Ace, a King, or any other card? Create a probability distribution in table and a histogram.
3. Construct a probability distribution for the number of tails when 2 balanced coins are tossed.

Problem No. 1

Step 1: List all possible outcomes.

Let the variable X represent the possible totals and the events are described as follows:

- E_1 is for students without failing subjects (80 students, thus, $n(E_1) = 80.$)
- E_2 is for students who fails in one subject (10 students, thus $n(E_2) = 10.$)
- E_3 is for students who fails in two subjects (7 students, thus $n(E_3) = 7.$)
- E_4 is for students who fails in three subjects (3 students, thus, $n(E_4) = 3.$)

Thus, the possible total (X) is:

$n(E_1)$	+	$n(E_2)$	+	$n(E_3)$	+	$n(E_4)$	=	Possible Total $n(E) = X$
80	+	10	+	7	+	3	=	$\boxed{100}$

Problem No. 1

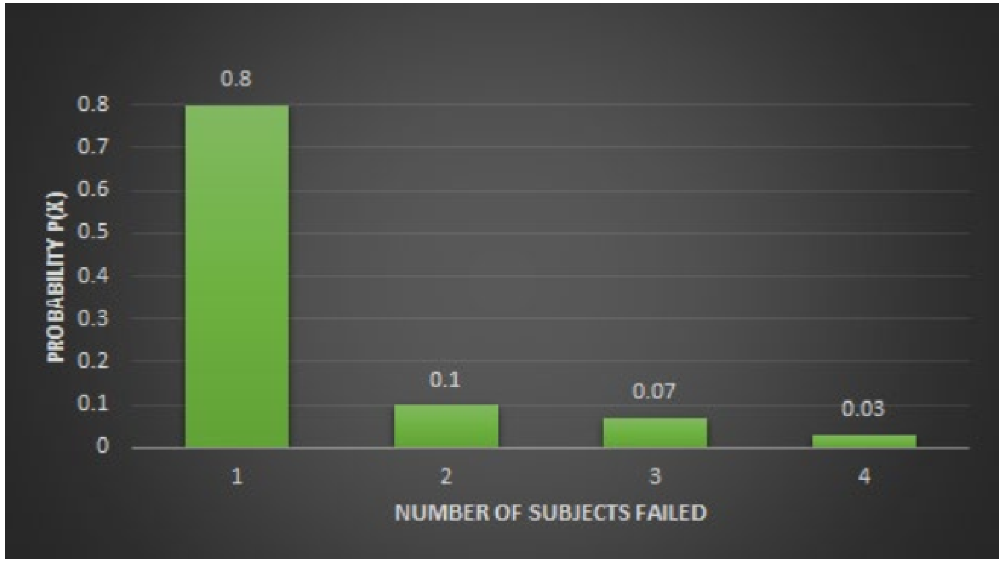
Steps 2 and 3: Identify outcomes that represent the same event. Find the probability of each event. Afterwards, Make a table or a histogram in which one column (or x-axis) represents the events and the other column (or y-axis) represents the probability.

Thus, the probability P(*X*) can be calculated for each *X* by dividing the number of subjects failed by the total number of students.

For failing in 0 subjects $F(0) = P(X = 0)$ $= f(0) = \frac{80}{100} \boxed{= 0.8}$	For failing in 1 subjects $F(1) = P(X = 1)$ $= f(10) = \frac{10}{100} \boxed{= 0.1}$
For failing in 2 subjects: $F(2) = P(X = 2)$ $= f(7) = \frac{7}{100} \boxed{= 0.07}$	For failing in 3 subjects: $F(3) = P(X = 3)$ $= f(3) = \frac{3}{100} \boxed{= 0.03}$

The probability distribution is shown below:

Number of subjects failed	0	1	2	3
Probability P(*X*)	0.8	0.1	0.07	0.03



Problem No. 2

Solution:

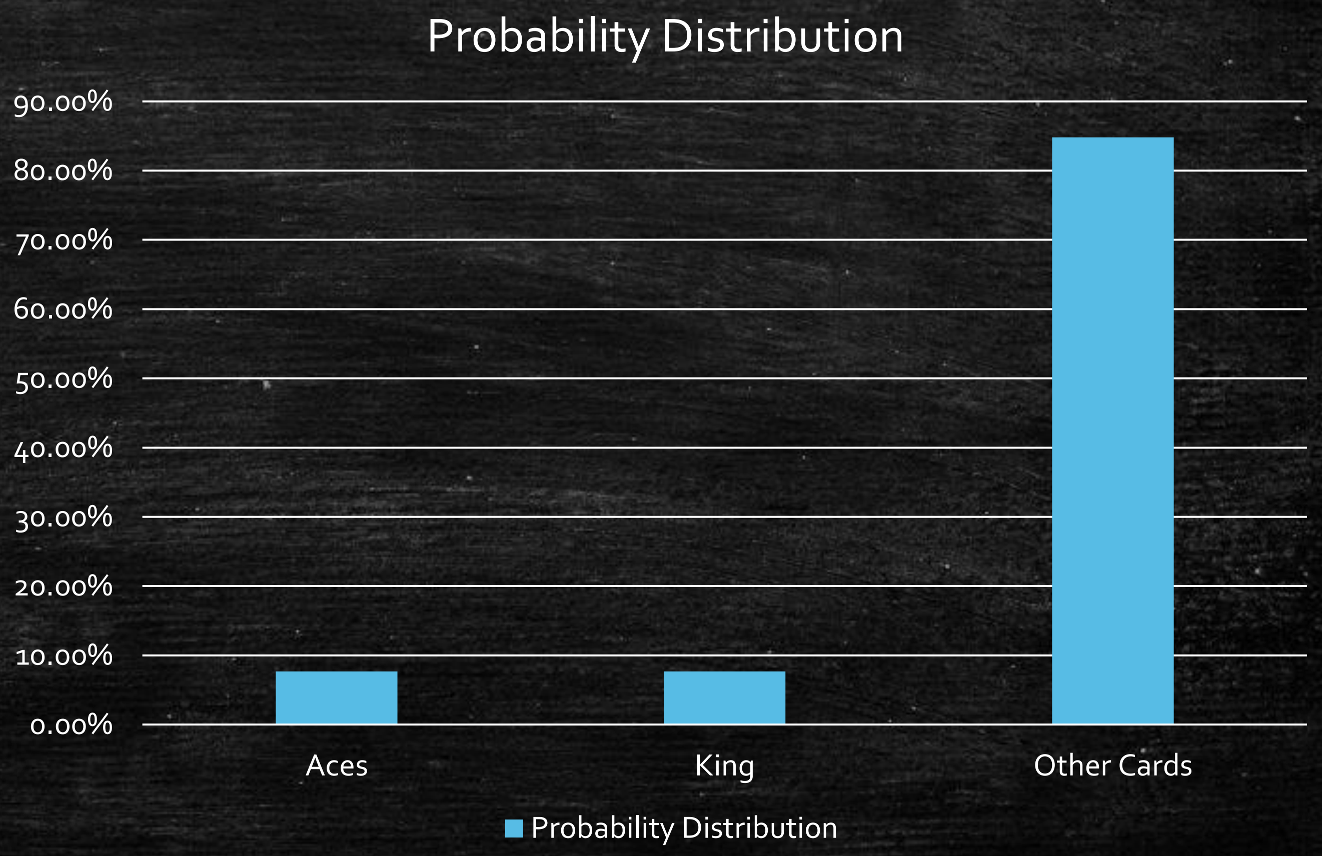
- Aces: 4
- Kings: 4
- Other cards: 52 - 4 - 4 = 44

Card Type	Probability
Ace	4/52 = 0.077
King	4/52 = 0.077
Other card	44/52 = 0.846

Problem No. 2

Probability Distribution

Cards	Ace	King	Other Cards
Probability	7.69%	7.69%	84.81%



Problem No. 3

Step 1: List all possible outcomes when tossing 2 coins

- HH (0 tails)
- HT (1 tail)
- TH (1 tail)
- TT (2 tails)

Thus, the possible total (X) is:

# of ways a 1st coin can land $n(E_1)$	$\times$	# of ways a 2nd coin can land $n(E_2)$	=	Possible Total $n(E) = X$
2	$\times$	**2**	**=**	**4**

Problem No. 3

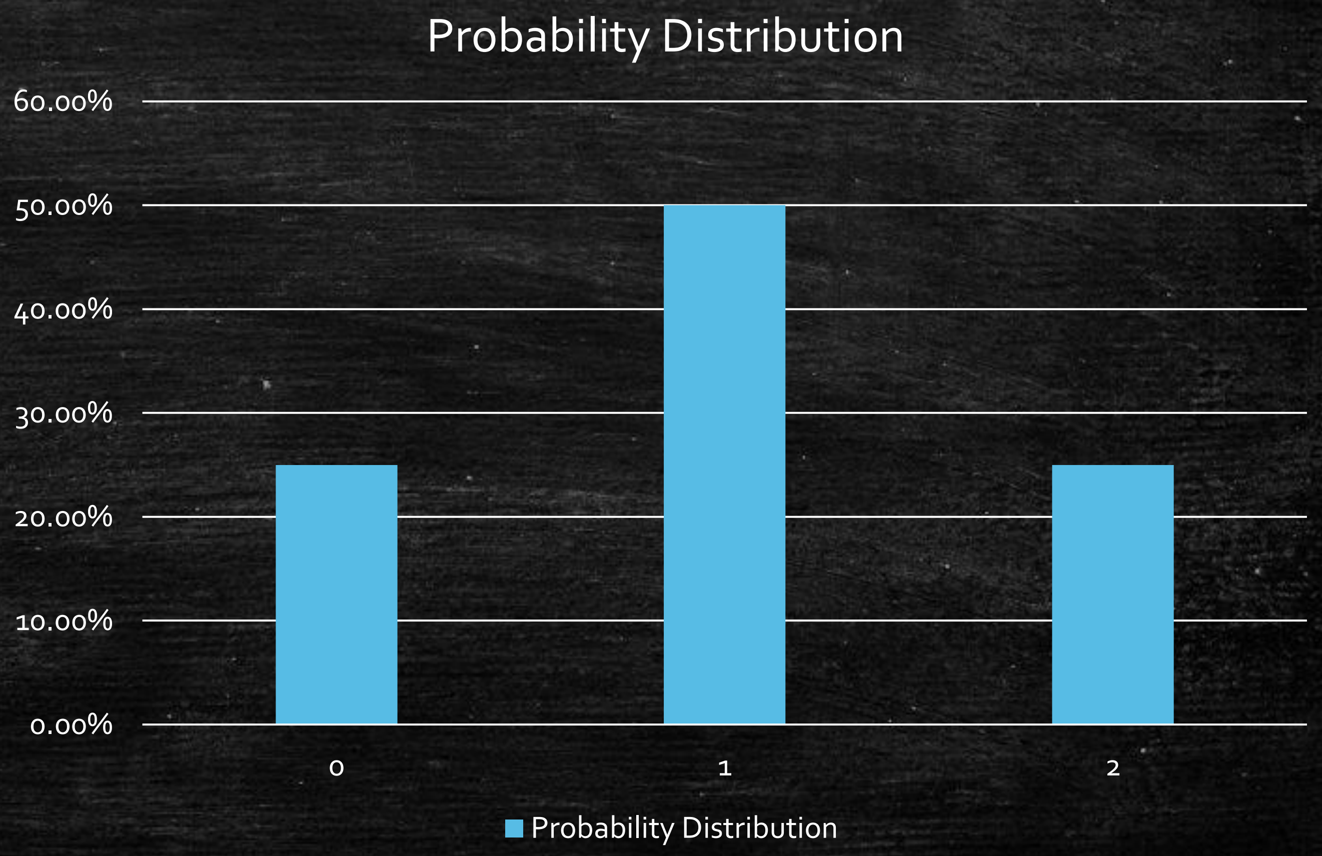
Step 2: Count the number of tails for each outcome and find probabilities

Number of tails (X)	Outcomes	Probability $P(X)$
0	HH	$\frac{1}{4} = 0.25$
1	HT, TH	$\frac{2}{4} = 0.5$
2	TT	$\frac{1}{4} = 0.25$

Problem No. 3

Probability Distribution

Number of Tails	0	1	2
Probability	25%	50%	25%



ACTIVITY:

13 5 1 14,

22 1 18 9 1 14 3 5,

19 20 1 14 4 1 18 4

4 5 22 9 1 20 9 15 14

ACTIVITY:

13 5 1 N ,

22 1 18 9 1 N 3 5 ,

19 20 1 N 4 1 18 4

4 5 22 9 1 20 9 15 N

ACTIVITY:

13 5 A N ,

22 A 18 9 A N 3 5 ,

19 20 A N 4 A 18 4

4 5 22 9 A 20 9 15 N

ACTIVITY:

M E A N ,

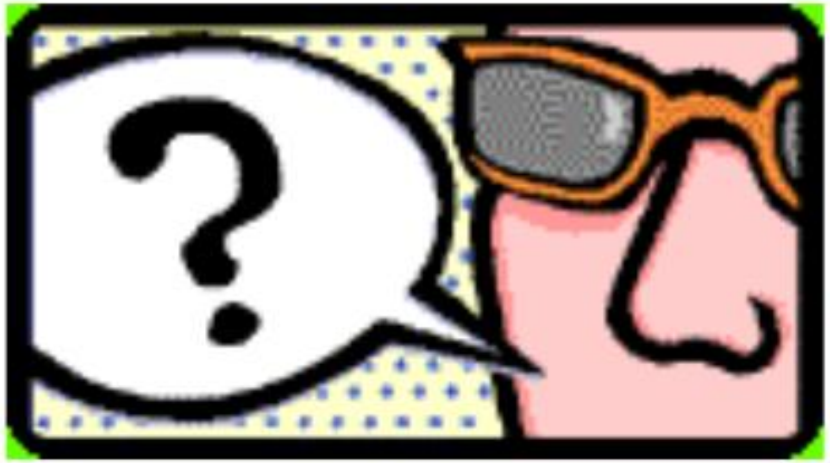
V A R I A N C E ,

S T A N D A R D

D E V I A T I O N

Expected Values of Random Variables

WHAT IS THE EXPECTED VALUE OF A RANDOM VARIABLE?



$$\mu_x = E(X) = \sum [x \cdot P(X = x)]$$

Expected Values of Random Variables

- **Variance:**

$$\delta^2 = \sum (X - \mu_x)^2 \cdot P(X = x)$$

- **Standard Deviation:**

$$\boldsymbol{\delta = \sqrt{\delta^2}}$$

📌 1. Mean (Average)

What is it?

The mean is the **average value** of a data set.

How to find it?

Add all the numbers and divide by how many numbers there are.

Formula:

$$\text{Mean} = \frac{\text{Sum of values}}{\text{Number of values}}$$

Example:

Scores: 2, 4, 6, 8

Mean = (2 + 4 + 6 + 8) ÷ 4 = 20 ÷ 4 = **5**

📌 2. Variance

What is it?

Variance measures how **spread out** the numbers are from the mean.

How to find it?

1. Subtract the mean from each value (find the deviation).
2. Square each deviation.
3. Find the average of those squared deviations.

Formula:

$$\text{Variance} = \frac{\sum(x - \text{mean})^2}{n}$$

Variance

Score		Deviations	Square of each Deviations	
2	Mean = 5	-3	9	Variance = 20/4= 5
4		-1	1	
6		1	1	
8		3	9	

Standard Deviation

📌 3. Standard Deviation

What is it?

The standard deviation is the **square root of the variance.**

It tells how much the values **typically differ** from the mean.

Formula:

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

Example:

Variance = 5

Standard Deviation = √5 ≈ **2.24**

Expected Values of Random Variables

Example 1:

Retrieved from: https://www.youtube.com/watch?v=JKGdEAEWUU8

No. of iPhones Sold (X)	0	1	2	3	4	5
Probability $P(x)$	0.05	0.10	0.12	0.15	0.18	0.40

No. of iPhones Sold (x)	0	1	2	3	4	5
Probability $P(X = x)$	0.05	0.10	0.12	0.15	0.18	0.40
$x \cdot P(X = x)$	0	0.10	0.24	0.45	0.72	2

Example 1

# of iPhone Sold (X)	Subtract (-)	Deviations $x - u_x$	Square of each Deviations $(x - u_x)^2$	Probability $P(x)$	$(x - u_x)^2 \cdot P(X=x)$	
0		-3.51	12.3201	.05	0.616005	
1		-2.51	6.3001	.10	0.63001	Variance $(\delta2)$ = 2.4899
2	Uniform Distribution (u_x) = 3.51	-1.51	2.2801	.12	0.273612	$\sqrt{\ }$
3		-0.51	.2601	.15	0.039015	
4		0.49	.2401	.18	0.043218	Standard Deviation (δ) = 1.57794
5		1.49	2.2201	.40	0.88804	

SELF CHECK:

Solve

Practice Exercise 2

of 03 Practice Exercises 1

03 PE2

Instruction: Find the expected value, variance, and standard deviation of the following probability distributions. Show the complete solution. (2 items x 10 points)

1.

x	0	1	2	3	4	5
$P(X = x)$	0.06	0.58	0.22	0.10	0.03	0.01

2.

x	0	1	2	3
$P(X = x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

03 PE2 – Problem No. 1

Item No. (x)	0	1	2	3	4	5
Probability P (X =X)	0.06	0.58	0.22	0.10	0.03	0.01
x . P (X =X)	0	0.58	0.44	0.3	0.12	0.05

Uniform Distribution (u_x)= 1.49

03 PE2 – Problem No. 1

Item No. (X)	Subtract (-)	Deviations $X - u_x$	Square of each Deviations $(X - u_x)^2$	Probability $P(x)$	$(x - u_x)^2 \cdot P(X=x)$	
0	Uniform Distribution (u_x) = 1.49	-1.49	2.2201	0.06	0.133206	Variance $(\delta 2)$ = 0.8699
1		-0.49	0.2401	0.58	0.139258	
2		0.51	0.2601	0.22	0.057222	
3		1.51	2.2801	0.10	0.22801	Standard Deviation (δ) = 0.93268
4		2.51	6.3001	0.03	0.189003	
5		3.51	12.3201	0.01	0.123201	

03 PE2 – Problem No. 2

Item No. (x)	0	1	2	3
Probability P (X =X)	1/8	3/8	3/8	1/8
x . P (X =X)	0	0.375	0.75	0.375

Uniform Distribution (u_x)= 1.50

03 PE2 – Problem No. 2

Item No. (X)	Subtract (-)	Deviations $x - u_x$	Square of each Deviations $(x - u_x)^2$	Probability $P(x)$	$(x - u_x)^2 \cdot P(X=x)$	
0	Uniform Distribution (u_x) = 1.50	-1.50	2.25	0.125	0.28125	Variance $(\delta 2)$ = 0.75
1		-0.50	0.25	0.375	0.09375	
2		0.50	0.25	0.375	0.09375	√
3		1.50	2.25	0.125	0.28125	Standard Deviation (δ) = 0.866